Ex. No.:3	Intelligent Email Assistant	URK23CS1167
27.08.2026		

Aim:

To develop an Intelligent Email Assistant using Ollama that generates context-aware customer replies by analyzing previous email interactions.

Problem Statement:

Develop an AI-powered Email Assistant capable of understanding customer queries, retrieving similar historical conversations, generating professional responses, improving tone, summarizing conversations, and storing generated replies.

Software Requirements:

Software	Version
Python	3.13.5
Ollama	0.40.13
VS Code	1.144.0
Pandas	3.1.0
Sentence Transformers	6.2.0
Flask	3.1.3
LLM Model	Llama 3.2 (3B) via Ollama

Input Parameters:

Parameters	Description
Customer Name	Name of the customer. Example: John
Subject	Subject line of the email. Example: Delivery Delay
Email	Body of the customer email. Example: My package has not arrived yet.

System Architecture:

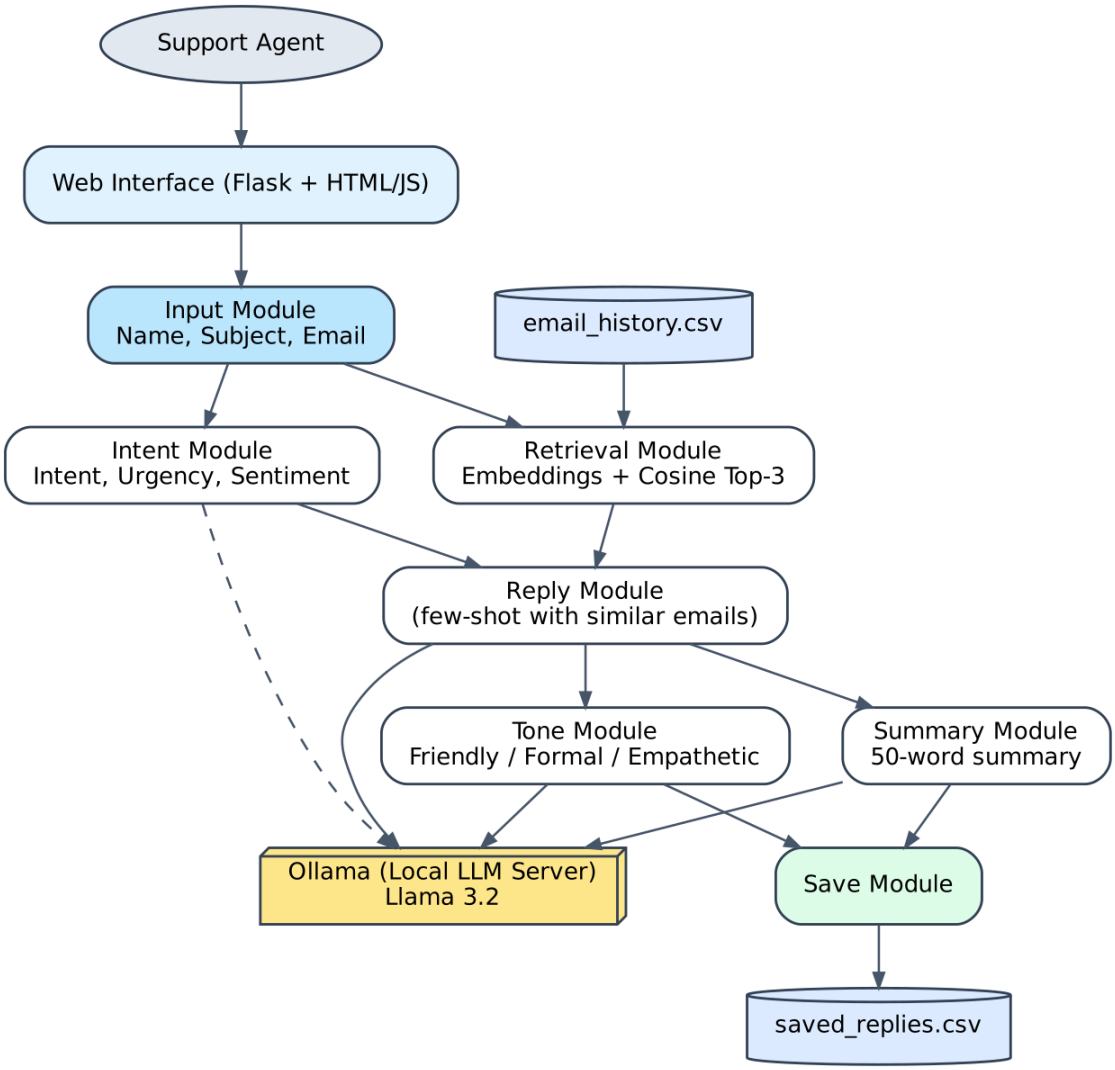


Fig. 3.1 System architecture

Module Design:

Module	Responsibility
Input Module	Collects customer name, subject and email body from the web form.
Intent Module	Uses the LLM to detect the intent (Delivery Delay, Refund, Damaged Product, etc.), urgency and sentiment of the email, with a keyword fallback.
Retrieval Module	Finds the top-3 most similar previous emails and their replies from email_history.csv using embeddings and cosine similarity.
Reply Module	Generates a professional reply using the new email and the retrieved conversations as few-shot examples.
Tone Module	Rewrites the reply in a Friendly, Formal or Empathetic tone and shows before/after.

Module	Responsibility
Summary Module	Generates a concise 50-word summary of the conversation for CRM notes.
Save Module	Saves the reply, intent, tone, summary and timestamp to saved_replies.csv.

Implementation:

• GitHub Link - https://github.com/MONICK10/GPT4-Exp3-Intelligent-Email-Assistant
• LinkedIn Post Link - <add LinkedIn post link in config.json>
• Run: `python app.py` → `http://localhost:5003`

Output Screenshots:

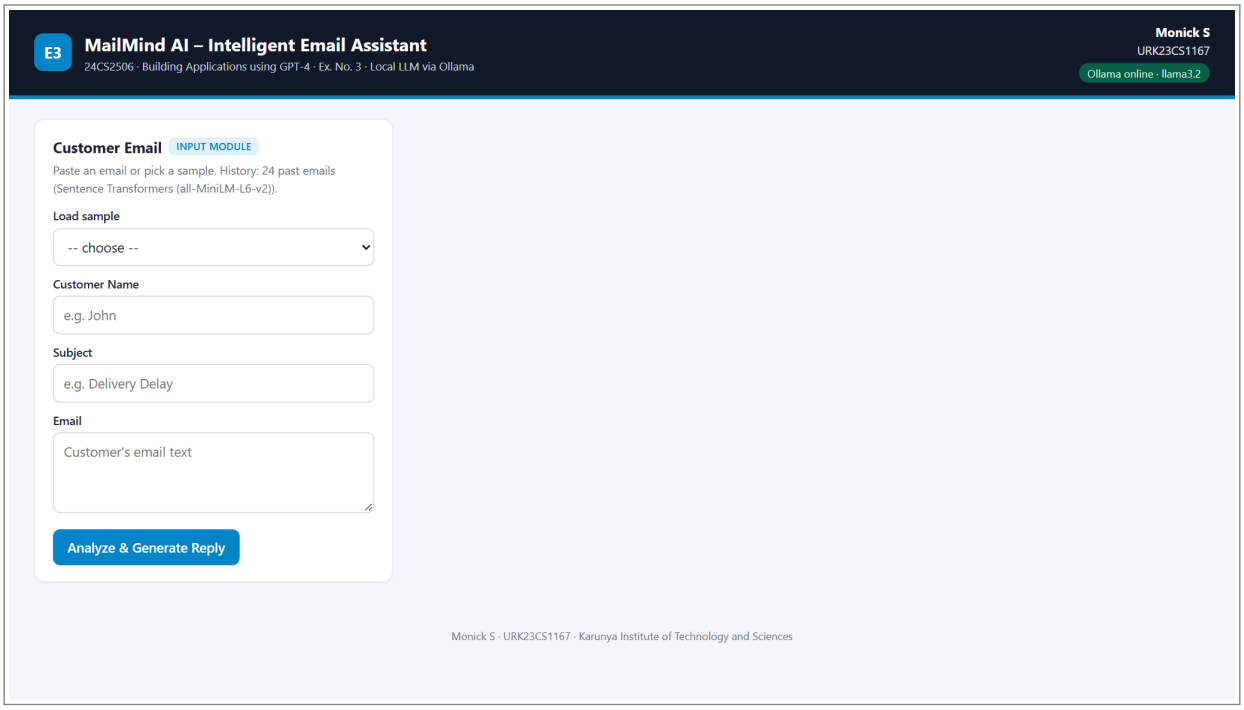


Fig. 3.2 Home Screen

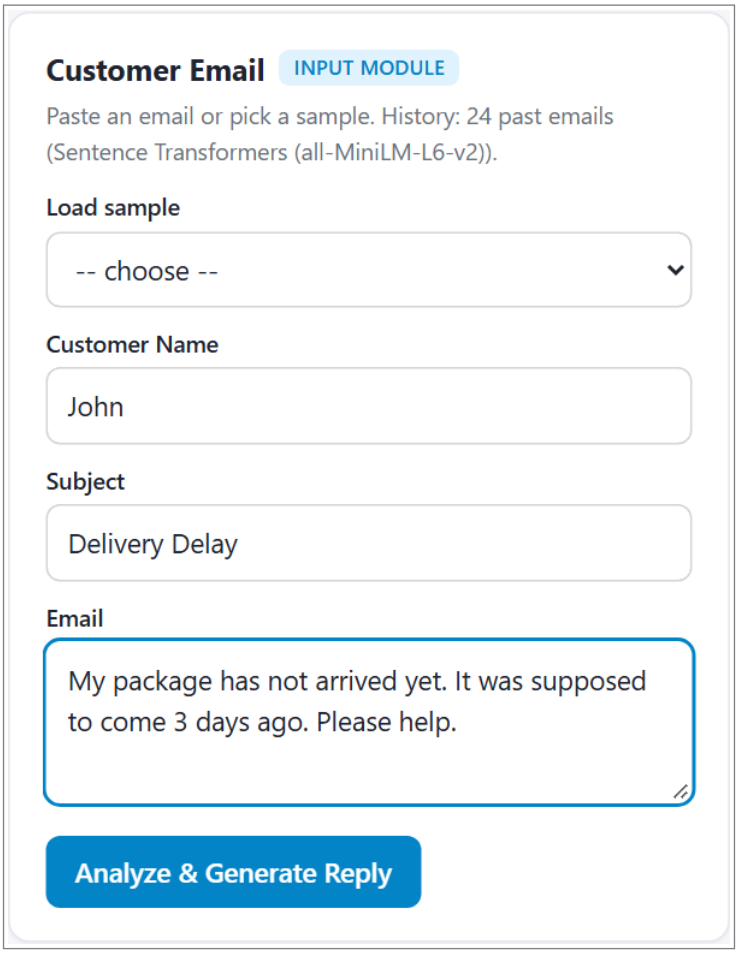


Fig. 3.3 Customer Email

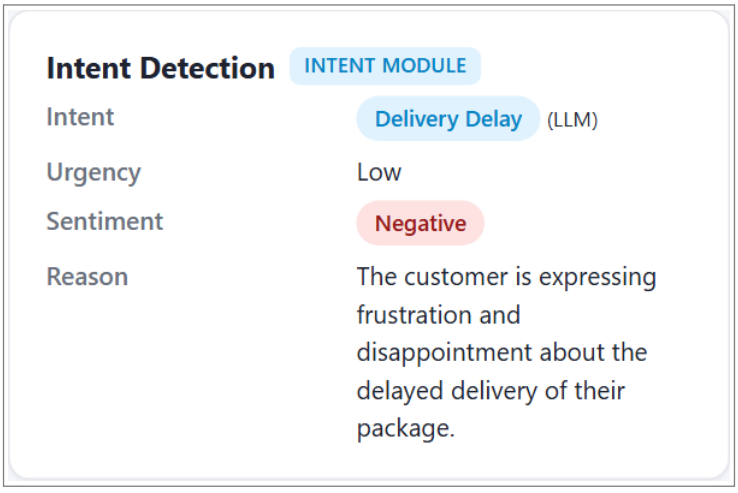


Fig. 3.4 Intent Detection

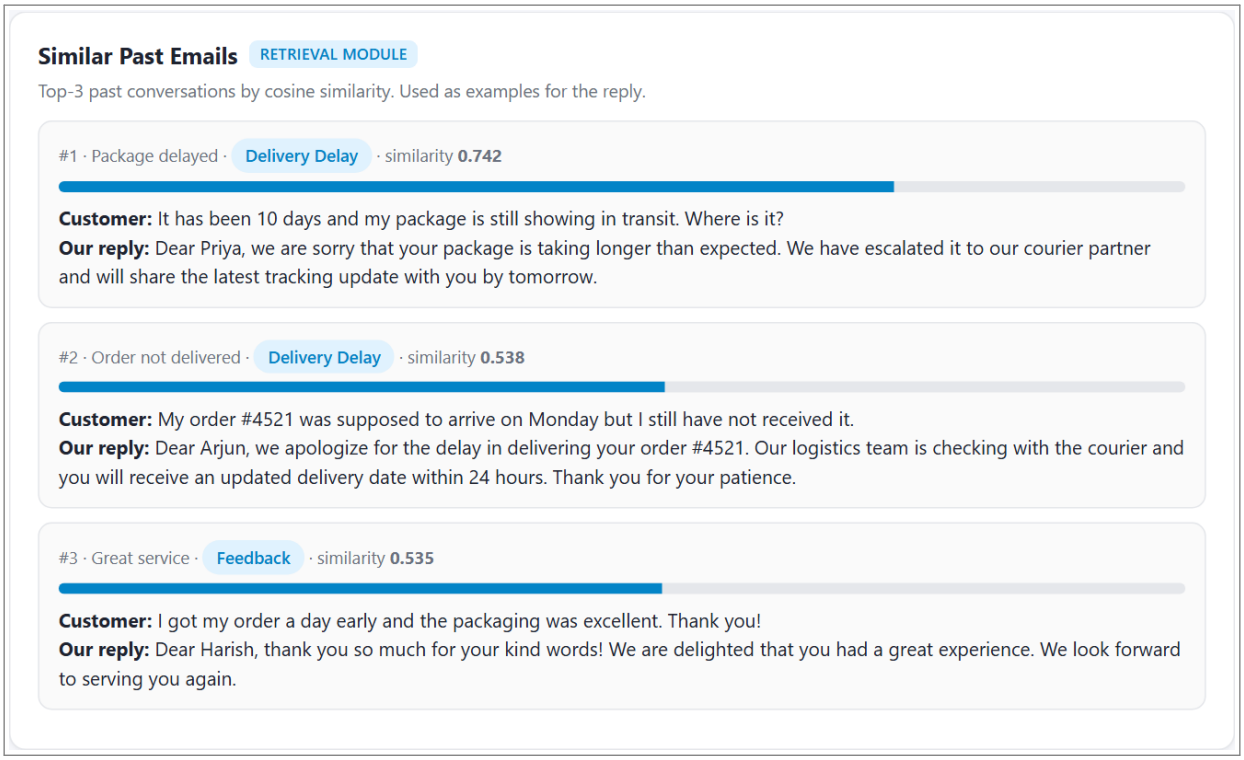


Fig. 3.5 Similar Email Retrieval

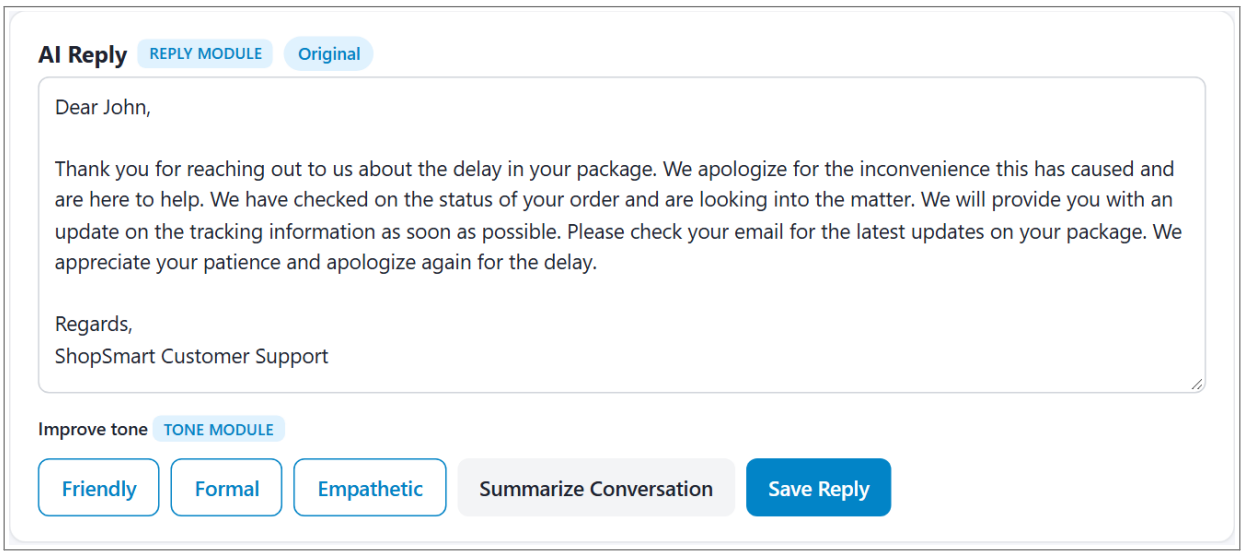


Fig. 3.6 AI Reply

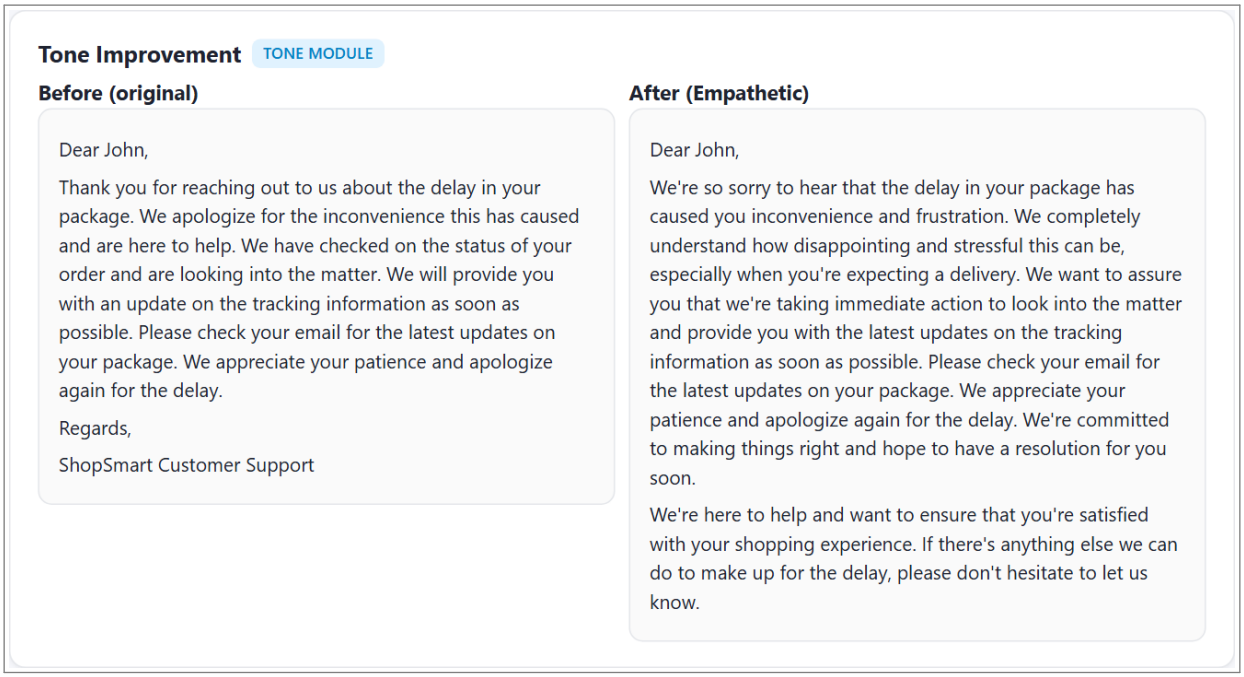


Fig. 3.7 Tone Improvement

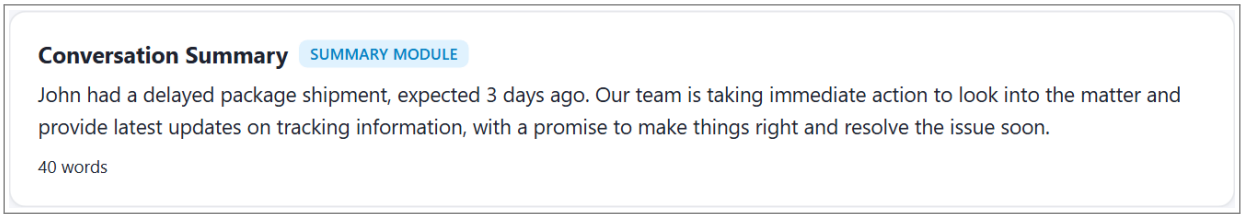


Fig. 3.8 Conversation Summary

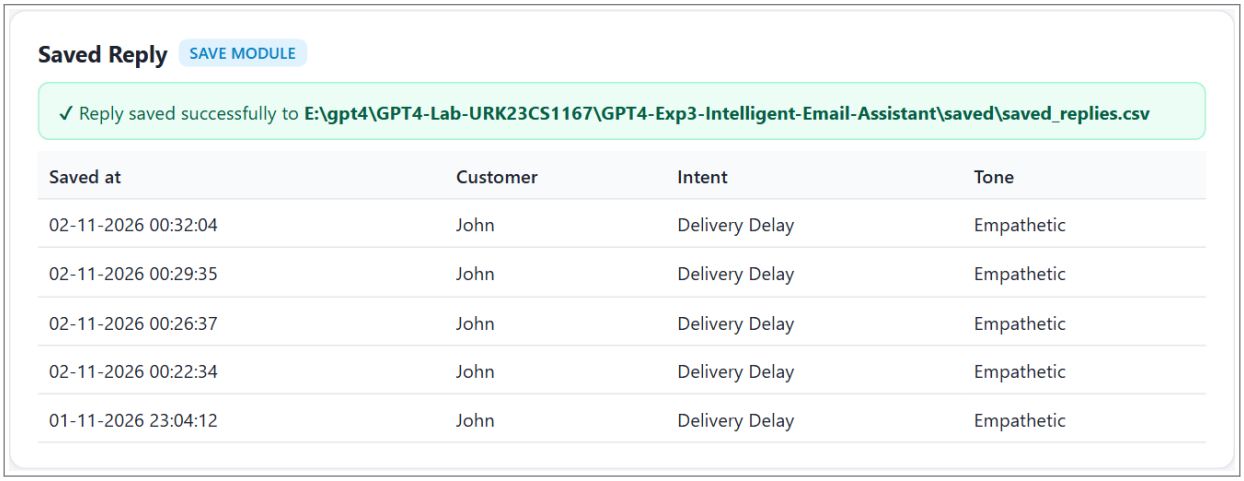


Fig. 3.9 Saved Reply

Result:

Thus, an Intelligent Email Assistant was successfully developed using Ollama. The application understands customer emails, retrieves similar historical conversations, generates professional responses, rewrites replies in different tones, summarizes conversations, and saves generated replies locally without requiring cloud-based APIs.